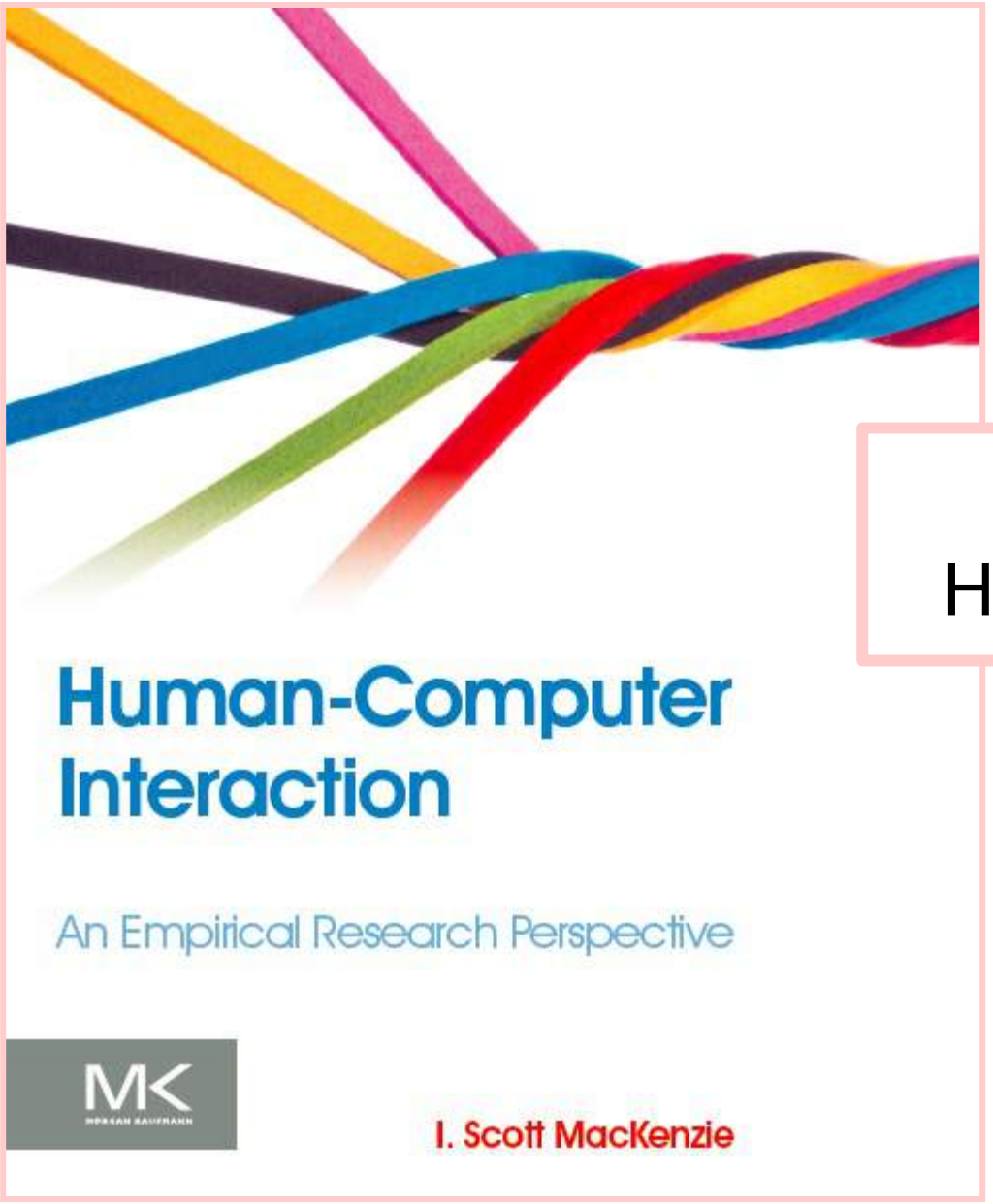
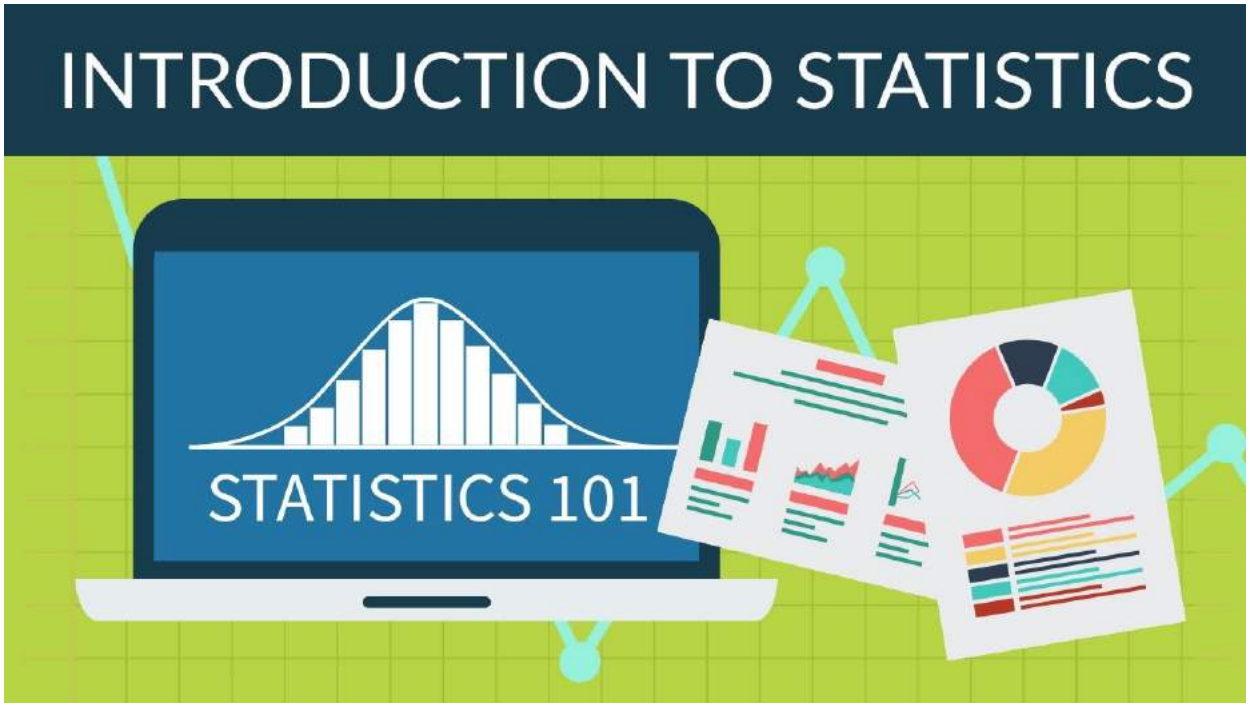


Chapter 6
Hypothesis Testing

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2018/04 Xiamen University

Warming Up

- [Introduction to Statistics]

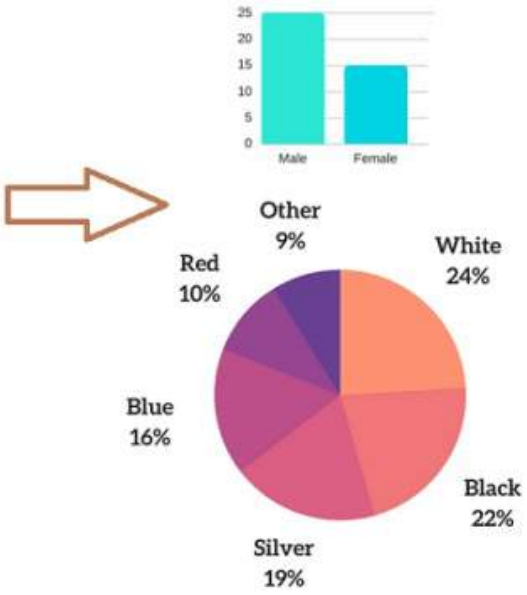


Practice

- [Excel Analysis ToolPak]
- [Descriptive Statistics]

	A	B	C	D
1	Respondent Number	Age	Gender	Favorite Car Color
2	1	22	M	White
3	2	37	F	Silver
4	3	45	F	Black
5	4	62	F	Gray
6	5	28	M	Red
7	6	45	M	Green
8	7	88	F	Brown
9	8	61	M	White
10	9	95	M	Black
11	10	27	M	White
12	11	39	F	Green
13	12	43	M	Brown
14	13	55	F	Black
15	14	59	F	White

RAW DATA



Descriptive Statistics

What is Hypothesis Testing?

- The use of statistical procedures to answer research questions
- Typical research question (generic):

> Is the time to complete a task less using Method A than using Method B?

- For hypothesis testing, hypothesis are always **statements**:

> There is no difference in the mean time to complete a task using Method A vs. Method B.

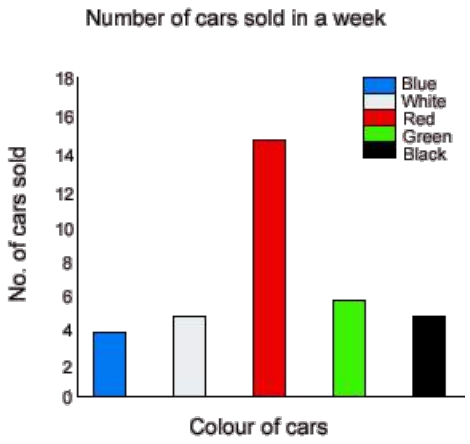
- This is the **null hypothesis** (assumption of "no difference")
- Statistical procedures seek to **reject** or **accept** the null hypothesis

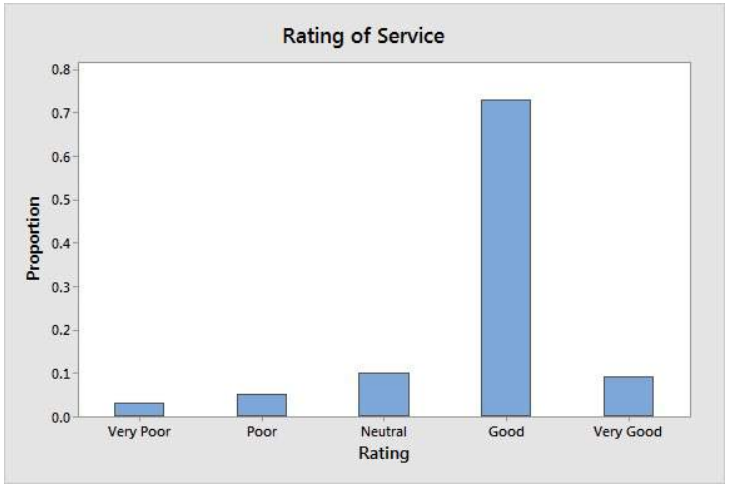
Statistical Procedures

- Two types:
  - Parametric
    - Data are assumed to come from a **distribution**, such as the normal distribution, *t*-distribution, etc.
  - Non-parametric
    - Data are not assumed to come from a distribution
  - Decision: match with the data types (measurement scale)
    - nominal, ordinal, interval, ratio

Measurement Scales vs. Statistical Tests

- Nominal
  - colour of cars: {Blue, White, Red, Green, Black}
- Ordinal
  - rating of service: {Very Poor, Poor, Neutral, Good, Very Good}





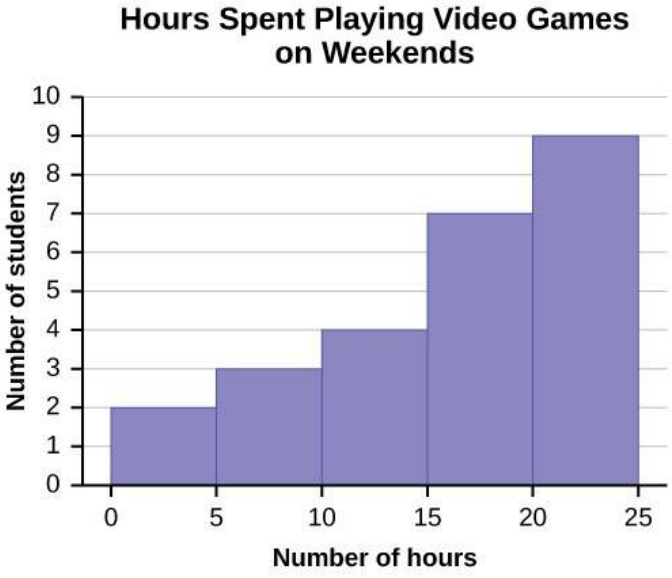
Measurement Scales vs. Statistical Tests

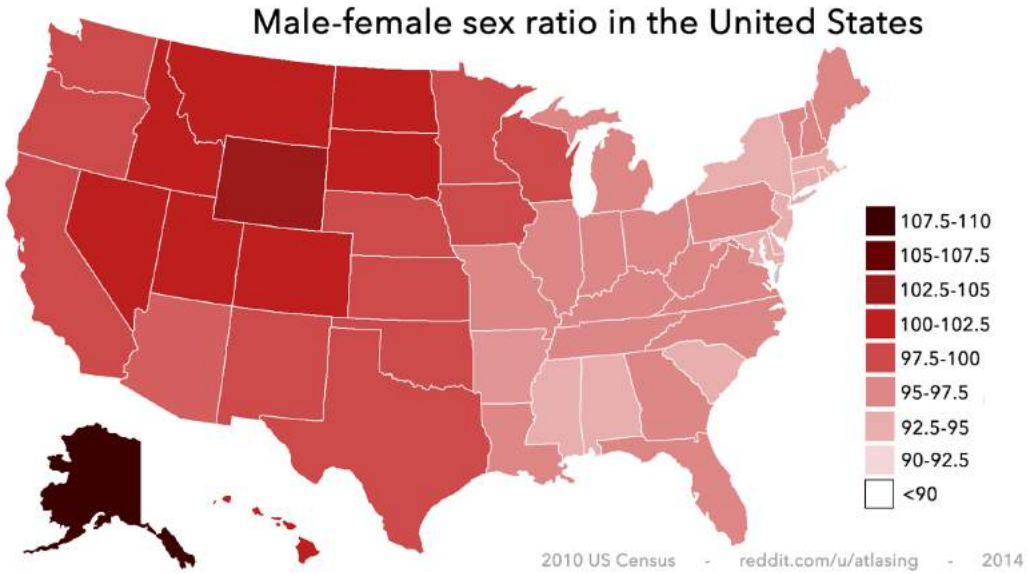
Measurement Scale	Defining Relations	Examples of Appropriate Statistics	Appropriate Statistical Tests
Nominal	• Equivalence	• Mode • Frequency	• Non-parametric tests
Ordinal	• Equivalence • Order	• Median • Percentile	
Interval	• Equivalence • Order • Ratio of intervals	• Mean • Standard deviation	• Parametric tests • Non-parametric tests
Ratio	• Equivalence • Order • Ratio of intervals • Ratio of values	• Geometric mean • Coefficient of variation	

- Non-parametric tests most appropriate for…
  - nominal data, ordinal data

Measurement Scales vs. Statistical Tests

- Interval
  - Hours on Video Games: [0:5:25]
- Ratio
  - Male-female sex ratio: [0.9--1.1]





Measurement Scales vs. Statistical Tests

Measurement Scale	Defining Relations	Examples of Appropriate Statistics	Appropriate Statistical Tests
Nominal	• Equivalence	• Mode • Frequency	• Non-parametric tests
Ordinal	• Equivalence • Order	• Median • Percentile	
Interval	• Equivalence • Order • Ratio of intervals	• Mean • Standard deviation	• Parametric tests • Non-parametric tests
Ratio	• Equivalence • Order • Ratio of intervals • Ratio of values	• Geometric mean • Coefficient of variation	

- Non-parametric tests most appropriate for…
  - nominal data, ordinal data,
- Parametric tests most appropriate for…
  - interval data, ratio data

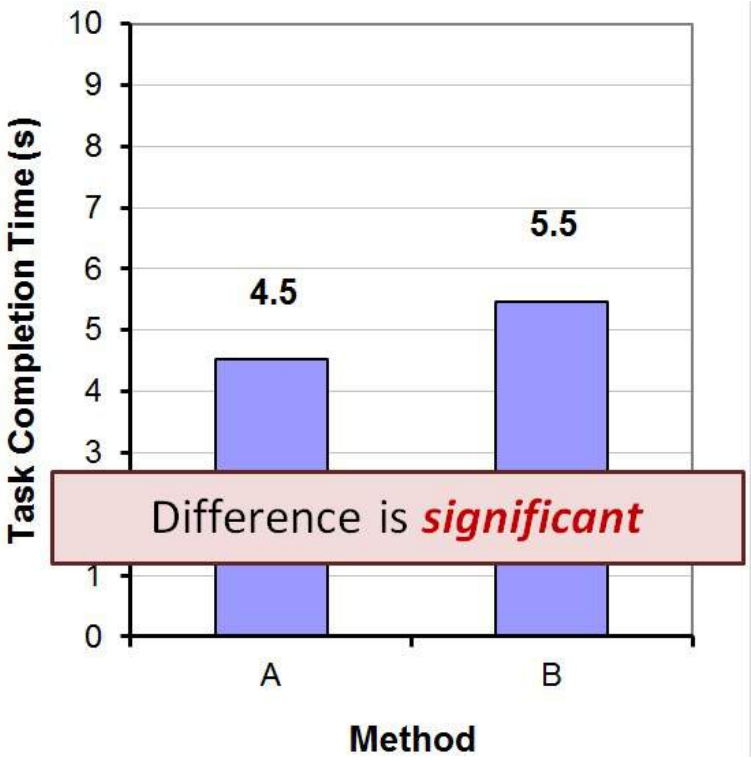
Tests Presented Here

- Parametric
  - Analysis of variance (**ANOVA**)
    - Used for ratio data and interval data
    - Most common statistical procedure in HCI research
- Non-parametric
  - Chi-square test
    - Used for nominal data
  - Mann-Whitney U, Wilcoxon Signed-Rank, Kruskal-Wallis, and Friedman tests
    - Used for ordinal data

Analysis of Variance

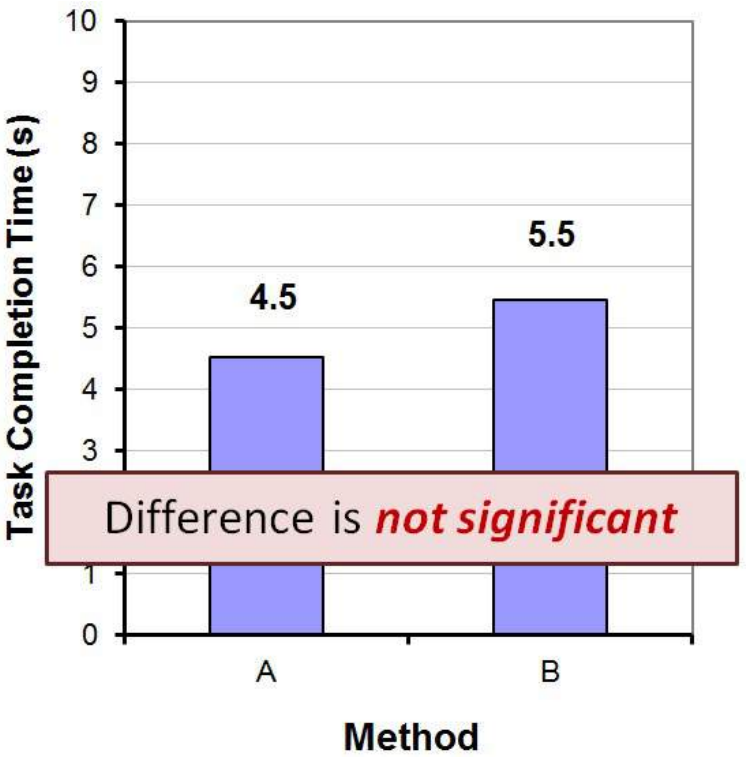
- **ANOVA:** most widely used statistical test for hypothesis testing
- Goal: determine if an independent variable has a **significant effect** on a dependent variable
- Goal: determine if the test conditions yield **different outcomes** on the dependent variable
  - e.g., one of the test conditions is faster than the other

Example #1



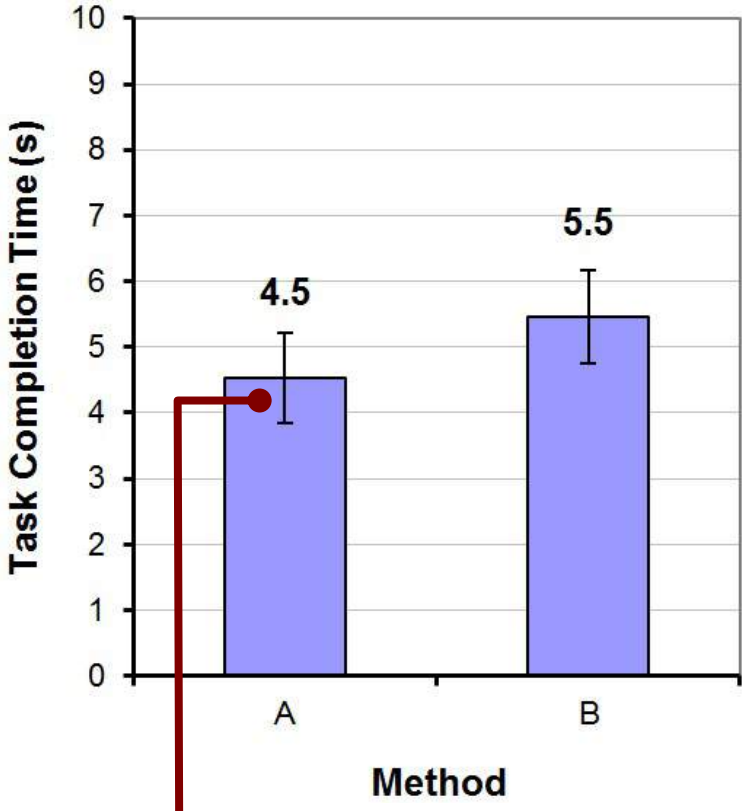
"Significant" implies that the difference observed is due to the test conditions (Method A vs. Method B).

Example #2



"Not significant" implies that the difference observed is likely due to chance (e.g., human errors)

Example #1 - Details



Note: Within-subjects design

Participant	Method	
	A	B
1	5.3	5.7
2	3.6	4.8
3	5.2	5.1
4	3.6	4.5
5	4.6	6.0
6	4.1	6.8
7	4.0	6.0
8	4.8	4.6
9	5.2	5.5
10	5.1	5.6
Mean	4.5	5.5
SD	0.68	0.72

Error bars show
±1 standard deviation

Note: *SD* is the square root of the variance

Example #1 – ANOVA[1]

ANOVA Table for Task Completion Time (s)

	DF	Sum of Squares	Mean Square	F-Value	P-Value	Lambda	Power
Subject	9	5.080	.564				
Method	1	4.232	4.232	9.796	.0121	9.796	.804
Method * Subject	9	3.888	.432				

Probability of obtaining the observed data if the null hypothesis is true

Reported as…

$F_{1,9} = 9.80, p < .05$

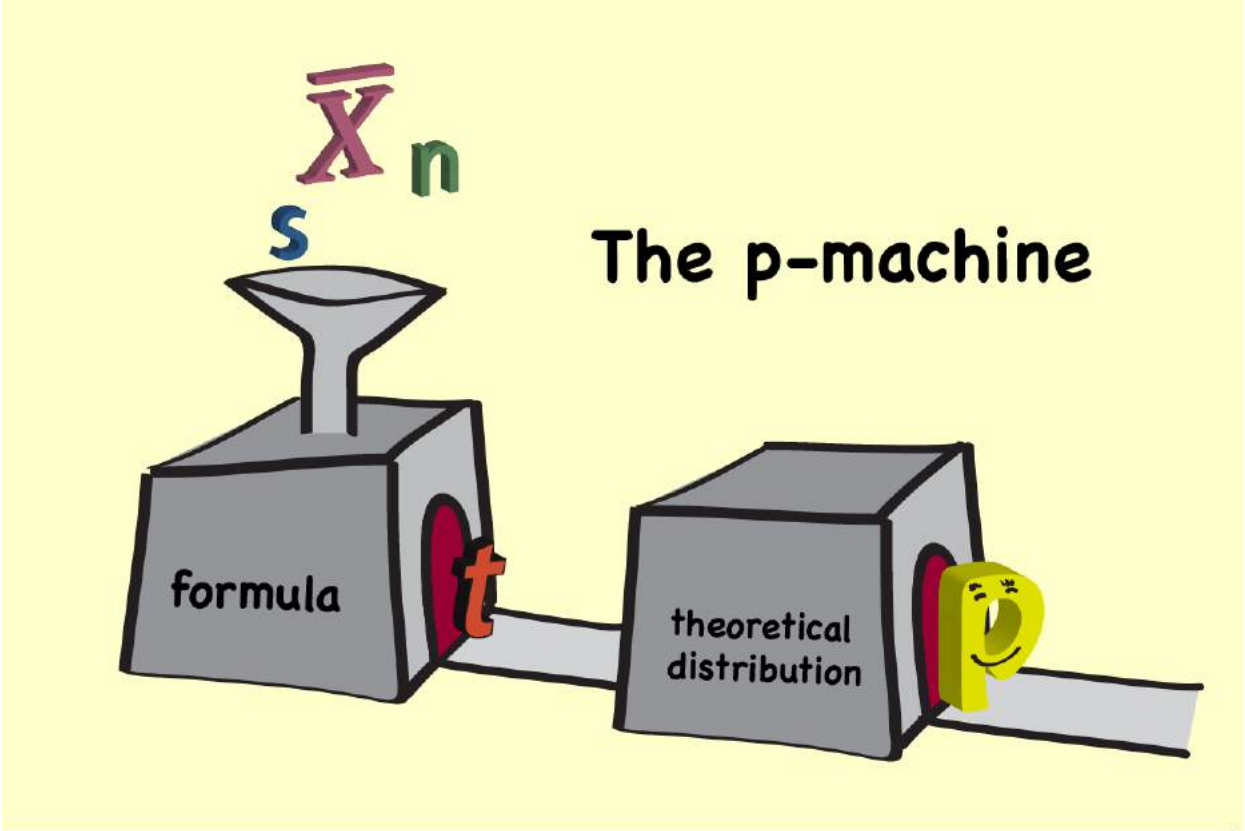
Thresholds for "p"

- .05
- .01
- .005
- .001
- .0005
- .0001

[1] ANOVA table created by *StatView* (now marketed as *JMP*, a product of SAS; www.sas.com)

Practice

- [Running ANOVA test in Excel]

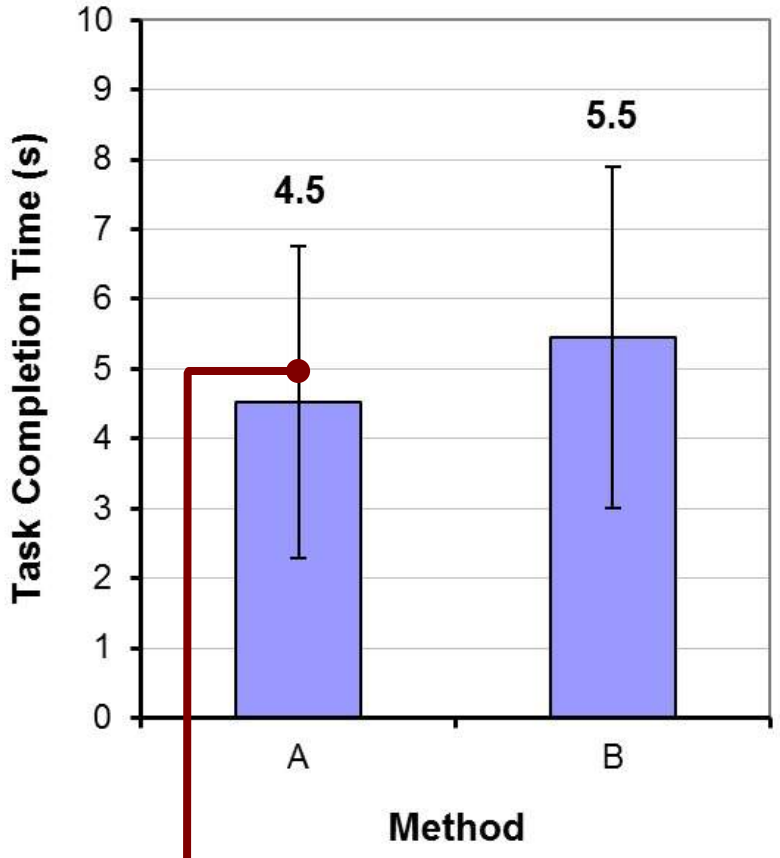


How to Report an *F*-statistic

> The mean task completion time for Method A was 4.5 s. This was 20.1% less than the mean of 5.5 s observed for Method B. The difference was statistically significant ($F_{1,9} = 9.80$, $p < .05$).

- Notice in the parentheses
  - Uppercase for *F*
  - Lowercase for *p*
  - Italics for *F* and *p*
  - Space both sides of equal sign
  - Space after comma
  - Space on both sides of less-than sign
  - Degrees of freedom are subscript, plain, smaller font
  - Three significant figures for *F* statistic
  - No zero before the decimal point in the *p* statistic (except in Europe)

Example #2 - Details



Error bars show
±1 standard deviation

Participant	Method	
	A	B
1	2.4	6.9
2	2.7	7.2
3	3.4	2.6
4	6.1	1.8
5	6.4	7.8
6	5.4	9.2
7	7.9	4.4
8	1.2	6.6
9	3.0	4.8
10	6.6	3.1
Mean	4.5	5.5
SD	2.23	2.45

Example #2 – ANOVA

ANOVA Table for Task Completion Time (s)

	DF	Sum of Squares	Mean Square	F-Value	P-Value	Lambda	Power
Subject	9	37.372	4.152				
Method	1	4.324	4.324	.626	.4491	.626	.107
Method * Subject	9	62.140	6.904				

Probability of obtaining the observed data if the null hypothesis is true

Note: For non-significant effects, use "**ns**"

Reported as…

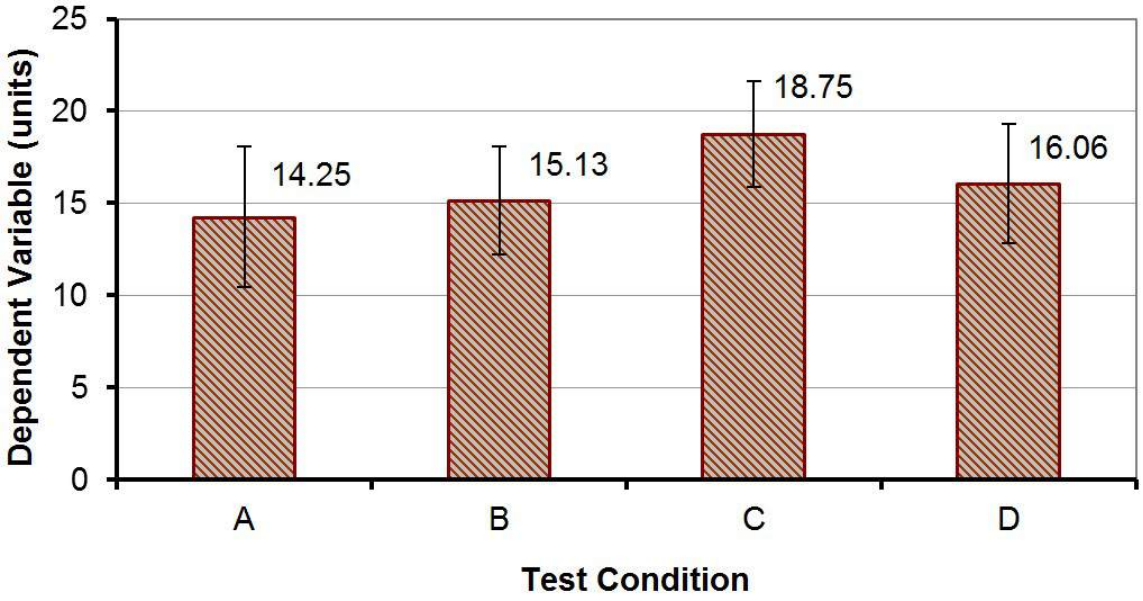
$F_{1,9} = 0.626$, **ns**

Example #2 - Reporting

The mean task completion times were 4.5 s for Method A and 5.5 s for Method B. As there was substantial variation in the observations across participants, the difference was not statistically significant as revealed in an analysis of variance ($F_{1,9}$ = 0.626, ns).

More Than Two Test Conditions

Participant	Test Condition			
	A	B	C	D
1	11	11	21	16
2	18	11	22	15
3	17	10	18	13
4	19	15	21	20
5	13	17	23	10
6	10	15	15	20
7	14	14	15	13
8	13	14	19	18
9	19	18	16	12
10	10	17	21	18
11	10	19	22	13
12	16	14	18	20
13	10	20	17	19
14	10	13	21	18
15	20	17	14	18
16	18	17	17	14
Mean	14.25	15.13	18.75	16.06
SD	3.84	2.94	2.89	3.23



ANOVA

ANOVA Table for Dependent Variable (units)

	DF	Sum of Squares	Mean Square	F-Value	P-Value	Lambda	Power
Subject	15	81.109	5.407				
Test Condition	3	182.172	60.724	4.954	.0047	14.862	.896
Test Condition * Subject	45	551.578	12.257				

- There was a significant effect of Test Condition on the dependent variable ($F_{3,45} = 4.95$, $p < .005$)
- Degrees of freedom
  - If n is the number of test conditions and m is the number of participants, the degrees of freedom are: $\mathbf{(n-1)(m-1)}$
  - Note: single-factor, within-subjects design

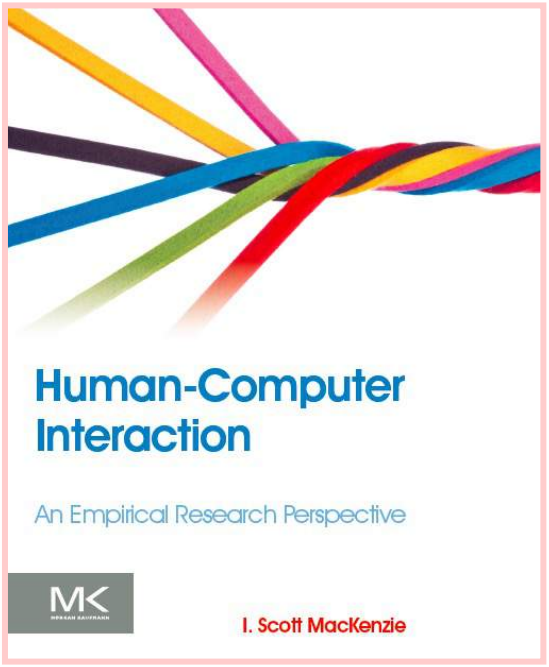
GoStats Software

- **HCI:ERP** web site includes GoStats download
- GUI app
- Replaces command-line apps demo'd in book
- Extensive API with demos, data files, discussions, etc.
- Data files for preceding examples
  - `anova-ex1.txt`
  - `anova-ex2.txt`
  - `posthoc-ex1.txt`
  - `betweensubjects.txt`
  - `twofactor.txt`





Thank You


Human-Computer Interaction
An Empirical Research Perspective
I. Scott MacKenzie